Half-Filled Sine-Kernel Endpoint and Extreme Modular Discrepancy

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Abstract

For the half-filled finite sine kernel and its commuting Jacobi matrix, we derive an exact integral formula and endpoint expansion for the reference scaling function, a Gaussian variational estimate for the largest Jacobi eigenvalue, and an exact Fourier-polynomial differential equation. The established fixed-index discrete-prolate concentration asymptotic determines the leading reference-modular discrepancy exactly, with limit $(\log \pi - 1)/2$. This is a fixed-mode conclusion and does not supply a uniform reference-modular estimate for the complete Wilson relative entropy.

1. Reference operators and conventions

Let $N = 2L$ and let the Hermitian $N \times N$ matrices B_N, J_N have entries

$$(B_N)_{jk} = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{i(j-k)\theta}\, d\theta, \qquad (J_N)_{j,j+1} = (J_N)_{j+1,j} = \frac{(j+1)(N-j-1)}{N},$$

for $0 \le j, k < N$, with all other entries of J_N zero. They commute, J_N has simple spectrum, and $0 < B_N < I$. Set

$$K_N = \log(I - B_N) - \log B_N, \qquad E_N = K_N + N\eta(J_N/N),$$

where

$$\eta(t) = \sum_{n=0}^{\infty} \frac{\pi \binom{2n}{n}\binom{4n}{2n}}{(2n+1)16^n} t^{2n+1}, \qquad |t| \le \tfrac{1}{2}.$$

The finite sine-kernel model specifies the reference operator and its commuting Jacobi operator.

2. Exact integral and endpoint expansion

Proposition 1. For $|t| \le 1/2$,

$$\boxed{\eta(t) = 4t \int_0^1 \frac{\sqrt{1-u^2}}{\sqrt{1-4t^2u^4}}\, du.} \qquad (2.1)$$

In particular,

$$\eta(1/2) = 2\,\mathrm{arsinh}\,1, \quad A_\eta := \max_{0<|t|\le 1/2} \frac{|\eta(t)|}{|t|} = 4\,\mathrm{arsinh}\,1,$$

and

$$C_\eta := \max_{0<|t|\le 1/2} \frac{|\eta(t)-\pi t|}{|t|^3} = 4(A_\eta - \pi).$$

Proof. Expand $(1-4t^2u^4)^{-1/2} = \sum_{n\ge 0}\binom{2n}{n}t^{2n}u^{4n}$. The beta integral gives

$$\int_0^1 u^{4n}\sqrt{1-u^2}\,du = \frac{\pi\binom{4n}{2n}}{4\,16^n(2n+1)},$$

which gives the stated coefficients. At the endpoint the majorant is integrable; monotone convergence applies to positive coefficients (the negative endpoint follows by oddness). For $t = 1/2$, cancellation of $1-u^2$ against $1-u^4$ gives $2\int_0^1(1+u^2)^{-1/2}\,du$. The two maxima occur at the endpoint because the respective power series have nonnegative coefficients. □

Theorem 2. As $\delta \downarrow 0$,

$$\boxed{\eta'(1/2-\delta) = \frac{1}{\sqrt{2}}\log\frac{1}{\delta} + 2\sqrt{2}\log 2 + o(1),} \qquad (2.2)$$

$$\boxed{\eta(1/2-\delta) = 2\,\mathrm{arsinh}\,1 - \frac{\delta}{\sqrt{2}}\log\frac{1}{\delta} - \frac{1+4\log 2}{\sqrt{2}}\,\delta + o(\delta).} \qquad (2.3)$$

Proof. For $t < 1/2$ differentiation under the integral gives

$$\eta'(t) = 4\int_0^1 \frac{\sqrt{1-u^2}}{(1-4t^2u^4)^{3/2}}\,du.$$

Set $\varepsilon = 1-4t^2$, $y = 1-u^2$. Then

$$\eta'(t) = 2\int_0^1 \frac{\sqrt{y}\,dy}{\sqrt{1-y}\,[\varepsilon + (1-\varepsilon)(2y-y^2)]^{3/2}}.$$

Subtract the singular model

$$I_\varepsilon = 2\int_0^1 \frac{\sqrt{y}}{(\varepsilon+2y)^{3/2}}\,dy = \sqrt{2}\left[\mathrm{arsinh}\,\sqrt{2/\varepsilon} - \sqrt{\frac{2}{2+\varepsilon}}\right].$$

Its asymptotic is

$$I_\varepsilon = \frac{1}{\sqrt{2}}\log\frac{1}{\varepsilon} + \frac{3}{\sqrt{2}}\log 2 - \sqrt{2} + o(1).$$

Subtracting the two integrands before sending $\varepsilon \to 0$ leaves an integrable function, uniformly bounded at $y = 0$: expand the smooth factors near $y = 0$ and use $y^2 + \varepsilon y \lesssim (\varepsilon + y)^2$. The endpoint $y = 1$ is integrable. Dominated convergence therefore gives the limiting difference

$$R = 2\int_0^1 \frac{1}{y}\left[\frac{1}{\sqrt{1-y}(2-y)^{3/2}} - \frac{1}{2^{3/2}}\right]dy.$$

With $u = \sqrt{1-y}$, this becomes

$$R = \int_0^1 \left[\frac{4}{(1-u^2)(1+u^2)^{3/2}} - \frac{\sqrt{2}u}{1-u^2} \right] du.$$

The antiderivative, zero at $u = 0$, is

$$2\frac{u}{\sqrt{1+u^2}} + \sqrt{2}\,\operatorname{artanh}\frac{\sqrt{2}u}{\sqrt{1+u^2}} + \frac{1}{\sqrt{2}}\log(1-u^2).$$

The logarithms cancel at $u = 1$, giving $R = \sqrt{2} + (3/\sqrt{2})\log 2$. Consequently

$$\eta'(t) = \frac{1}{\sqrt{2}}\log\frac{1}{\varepsilon} + 3\sqrt{2}\log 2 + o(1).$$

Since $t = 1/2 - \delta$ entails $\varepsilon = 4\delta(1-\delta)$, this is (2.2). Integrating the derivative between t and $1/2$, and using the endpoint value, yields (2.3). □

3. Gaussian comparison for the top Jacobi eigenvalue

Theorem 3. Let λ_N be the maximum eigenvalue of J_N for even N. Then

$$\boxed{\lambda_N = \frac{N}{2} - \frac{1}{\sqrt{2}} + O(N^{-1}).} \qquad (3.1)$$

In addition, writing $\beta = \sqrt{2}$, every even $N \geq 2$ obeys the explicit bound

$$\boxed{\lambda_N \leq e^{-\beta/N}\left(\frac{N}{2} + \frac{6\cosh^2\beta}{5N}\right).} \qquad (3.2)$$

Proof. Write $x_j = j - (N-1)/2$ and choose $w_j = e^{-\beta x_j^2/N} > 0$. Collatz–Wielandt gives $\lambda_N \leq \max_j (J_N w)_j / w_j$. An exact calculation with $y = 2\beta x_j/N$ yields

$$\frac{(J_N w)_j}{w_j} = e^{-\beta/N}\left[\frac{N}{2}\left(1 - \frac{y^2}{2}\right)\cosh y - \frac{\cosh y}{2N} + \frac{y}{\beta}\sinh y\right].$$

The power series of $(1 - y^2/2)\cosh y$ has no quadratic term, a quartic coefficient $-5/24$, and strictly negative higher even coefficients, so the expression is at most $1 - 5y^4/24$. Also $|y| \leq \beta$ and $y\sinh y \leq y^2\cosh\beta$. Hence

$$\frac{(J_N w)_j}{w_j} \leq e^{-\beta/N}\left[\frac{N}{2} - \frac{5N}{48}y^4 + \frac{\cosh\beta}{\beta}y^2\right] \leq e^{-\beta/N}\left[\frac{N}{2} + \frac{6\cosh^2\beta}{5N}\right],$$

where the final step maximizes the displayed quadratic polynomial in $y^2 \geq 0$.

For the lower bound use the Rayleigh quotient of w. With $z_j = j + 1 - N/2$ for the bonds, an exact expression is

$$R_N = 2e^{-\beta/(2N)}\frac{\sum_{j=0}^{N-2}(N/4 - z_j^2/N)e^{-2\beta z_j^2/N}}{\sum_{j=0}^{N-1} e^{-2\beta x_j^2/N}} \leq \lambda_N.$$

Both sums are truncated Gaussian lattice sums. Extending to the full integer and half-integer lattices incurs exponentially small errors (up to polynomial prefactors), since the first omitted points have magnitude proportional to N. Poisson summation then gives

$$\sum_{k\in\mathbb{Z}} e^{-2\beta(k+\rho)^2/N} = \sqrt{\frac{\pi N}{2\beta}}\left[1 + O\left(e^{-\pi^2 N/(2\beta)}\right)\right], \qquad \rho \in \{0, 1/2\},$$

and the corresponding second Gaussian moment is $N/(4\beta)$ times the zeroth moment, up to exponentially small absolute relative errors. Therefore, for some $c > 0$,

$$R_N = e^{-\beta/(2N)}\left(\frac{N}{2} - \frac{1}{2\beta}\right) + O(e^{-cN}) = \frac{N}{2} - \frac{1}{\sqrt{2}} + \frac{3}{8N} + O(N^{-2}).$$

Together with (3.2), which equals $N/2 - 1/\sqrt{2} + O(N^{-1})$, this proves (3.1). The $3/(8N)$ coefficient in the *lower bound* is not being asserted as the next coefficient of the true eigenvalue. □

Corollary 4. Let $\lambda_N = \lambda_{\max}(J_N)$. Then

$$\boxed{N\eta(\lambda_N/N) = 2N \operatorname{arsinh} 1 - \frac{1}{2}\log N - \frac{1 + \log(16\sqrt{2})}{2} + o(1).} \qquad (3.3)$$

Proof. Set $\delta_N = 1/2 - \lambda_N/N = 1/(\sqrt{2}N) + O(N^{-2})$ in Theorem 2. The $O(N^{-2})$ input error gives $o(1)$ after multiplication by N, since η' grows only logarithmically. Simplification gives (3.3). □

4. An exact generating-polynomial and Fourier-leakage formulation

Theorem 5. Let $J_N v = \lambda v$ for a nonzero real vector $v = (v_0, \dots, v_{N-1})$ and define the polynomial $F(z) = \sum_{j=0}^{N-1} v_j z^j$. It satisfies the exact differential equation

$$\boxed{-z(1+z^2)F''(z) + [(N-1) + (N-3)z^2]F'(z) + [(N-1)z - N\lambda]F(z) = 0.} \qquad (4.1)$$

If $b = v^* B_N v/\|v\|^2$, then

$$\boxed{\frac{1-b}{b} = \frac{\int_{\pi/2<|\theta|\le\pi} |F(e^{i\theta})|^2\, d\theta}{\int_{-\pi/2}^{\pi/2} |F(e^{i\theta})|^2\, d\theta}.} \qquad (4.2)$$

For the top eigenpair, v may be chosen strictly positive and symmetric under $j \mapsto N-1-j$; consequently $F(z) = z^{N-1}F(1/z)$.

Proof. The recurrence $N\lambda v_j = (j+1)(N-j-1)v_{j+1} + j(N-j)v_{j-1}$ is understood with $v_{-1} = v_N = 0$. Multiply by z^j and sum over j. The shifted sums give $(N-1)F' - zF''$ and $(N-1)zF + (N-3)z^2F' - z^3F''$, respectively, proving (4.1). Equation (4.2) follows directly from the Fourier projection formula for B_N and Parseval. Positivity and strict Perron simplicity hold because all Jacobi off-diagonal entries are positive; reflection commutes with J_N, hence the top vector is symmetric. □

Fourier-leakage identity and endpoint criterion. Let v_N be the top normalized Jacobi eigenvector, and write I_N^{in} and I_N^{out} for the two integrals in (4.2). Its scalar modular discrepancy is

$$e_N^{\text{edge}} = \log\frac{I_N^{\text{out}}}{I_N^{\text{in}}} + N\eta(\lambda_N/N).$$

By (3.3), **boundedness of this one edge discrepancy** is equivalent to

$$\boxed{\frac{I_N^{\text{out}}}{I_N^{\text{in}}} = \sqrt{N}\, e^{-2N \operatorname{arsinh} 1 + O(1)}.} \qquad (4.3)$$

The $O(1)$ is in the logarithm and means two-sided multiplicative bounds independent of N. The fixed-index concentration asymptotic in the next section establishes this criterion for the leading mode. It does not provide uniform control over eigenvalue indices that grow with N.

5. Extreme modular discrepancy from classical discrete-prolate concentration

Theorem 6 (application of the classical discrete-prolate fixed-index asymptotic). In the preceding half-filled normalization,

$$\boxed{1 - b_N \sim \sqrt{\pi}\, 2^{9/4} \sqrt{N}\, e^{-2N \operatorname{arsinh}(1)}.} \qquad (5.1)$$

Consequently,

$$\boxed{\lim_{\substack{N\to\infty \\ N \text{ even}}} e_N = \frac{\log \pi - 1}{2}.} \qquad (5.2)$$

In particular, the discrepancy for the **single top Jacobi eigenmode** is bounded uniformly over even $N \geq 2$.

Proof. The classical discrete-prolate concentration matrix is

$$P(N, W)_{jk} = \begin{cases} \frac{\sin(2\pi W(j-k))}{\pi(j-k)}, & j \neq k, \\ 2W, & j = k. \end{cases}$$

It equals B_N for $W = 1/4$. The fixed-index eigenvalue asymptotic of Slepian (1978), explicitly transcribed in Adcock, Huybrechs and Martín-Vaquero, *On the numerical stability of Fourier extensions*, §3.2, equation (3.10), states for $k = 0$,

$$1 - b_N \sim \sqrt{\pi}\, 2^{9/4} \alpha^{1/4} (2-\alpha)^{-1/2} \sqrt{N}\, \beta^{-N}, \quad \alpha = 1 - \cos(2\pi W), \quad \beta = \frac{\sqrt{2} + \sqrt{\alpha}}{\sqrt{2} - \sqrt{\alpha}}.$$

At $W = 1/4$, $\alpha = 1$, while $\beta = 3 + 2\sqrt{2} = \exp(2 \operatorname{arsinh} 1)$. This proves (5.1). Since $b_N \to 1$, its odds satisfy

$$\log \frac{1 - b_N}{b_N} = -2N \operatorname{arsinh}(1) + \frac{1}{2} \log N + \log(\sqrt{\pi} 2^{9/4}) + o(1).$$

Adding the Jacobi-edge expansion (3.3) and simplifying the constants proves (5.2). A convergent sequence is bounded for sufficiently large N; the finitely many remaining even N have finite discrepancies since $0 < b_N < 1$. □

Source attribution. The concentration-eigenvalue asymptotic used here is due to Slepian and is reproduced by Adcock, Huybrechs and Martín-Vaquero. This is an application of their theorem, not an independent proof of it. The source's theorem applies to fixed eigenvalue index and fixed bandwidth; no uniformity in an index growing with N has been imported. The proof of Slepian's original asymptotic is not reproduced here.

6. Exact real angular equation

Let $J_N v = \lambda v$, $F(z) = \sum_{j=0}^{N-1} v_j z^j$, and $\phi(\theta) = e^{-i(N-1)\theta/2} F(e^{i\theta})$. Summing the Jacobi recurrence gives

$$-z(1+z^2)F'' + [(N-1) + (N-3)z^2]F' + [(N-1)z - N\lambda]F = 0.$$

Proposition 7. Every generating polynomial satisfies

$$\boxed{\frac{d}{d\theta}(\cos\theta\,\phi'(\theta)) + \left[\frac{N^2-1}{4}\cos\theta - \frac{N\lambda}{2}\right]\phi(\theta) = 0.} \qquad (6.1)$$

Proof. On the unit circle $z\partial_z = -i\partial_\theta$. Substitute $F(e^{i\theta}) = e^{i(N-1)\theta/2}\phi(\theta)$ into the displayed polynomial equation, collect terms in ϕ'', ϕ', and ϕ, and divide by two. The resulting expression is (6.1). This algebra is independent of any asymptotic approximation. □

For the top eigenvector, reflection symmetry makes ϕ real and even. For even N, palindromicity of odd polynomial degree implies $\phi(\pi) = 0$, and (6.1) at the band edge gives

$$\phi'(\pi/2) = -\frac{N\lambda_N}{2}\phi(\pi/2). \qquad (6.2)$$

Multiplying (6.1) by ϕ and integrating by parts gives the exact exterior identity

$$\boxed{\int_{\pi/2}^{\pi} |\cos\theta|\,\phi'(\theta)^2\,d\theta = \int_{\pi/2}^{\pi}\left[\frac{N^2-1}{4}|\cos\theta| + \frac{N\lambda_N}{2}\right]\phi(\theta)^2\,d\theta.} \qquad (6.3)$$

The analogous interior formula over $[0, \pi/2]$ has $\cos\theta$ instead of its absolute value and the minus sign preceding $N\lambda_N/2$. Neither identity supplies a two-sided exterior mass estimate by itself.

7. Exact action and asymptotic interpretation

The leading interior normal-form action computed from (6.1) is

$$\mathcal{A} = \frac{1}{2}\int_0^{\pi/2} \sqrt{\sec\theta - 1}\,d\theta = \operatorname{arsinh}(1).$$

Indeed, with $t = \tan(\theta/2)$ and then $u = \sqrt{1-t^2}$, the integral becomes $\sqrt{2}\int_0^1 (2-u^2)^{-1}du = \operatorname{artanh}(1/\sqrt{2}) = \operatorname{arsinh}(1)$.

This matches the exponential action in the classical theorem (5.1). It is an exact evaluation of a leading formal action, **not** a standalone rigorous derivation of the prolate eigenvalue prefactor. In particular, the behavior at the singular band edge and the central turning point requires controlled connection formulas if one seeks a direct ODE proof.

8. Scope and limitations

The fixed-index theorem controls only the highest Jacobi eigenvector and its paired sine-kernel occupation. It cannot be extended to an index growing with N without new estimates. The commuting sine-kernel/Jacobi spectrum includes central, transitional, and other edge regions that are not bounded by the single-mode result.

For the Wilson–Dirac interval problem the established reduction is

$$\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + \operatorname{Tr}(S_N E_N) + O_\mu(\sqrt{\log L}), \qquad N = 2L.$$

Neither the scalar endpoint expansion nor the Slepian fixed-index result alone establishes the leading coefficient or finite part of the complete relative entropy. The separate argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* gives $\operatorname{Tr}(S_N E_N) = O_\mu(1)$ and the leading entropy coefficient; it does not establish the entropy finite part or an expansion uniform up to the spectral edge.

References

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